

Singular Pooling: A Spectral Pooling Paradigm for Second-Trimester Prenatal Level II Ultrasound Standard Fetal Plane Identification

Shengjun Zhu¹, Student Member, IEEE, Jiaxin Cai¹, Runqing Xiong, Liping Zheng, and Duo Ma

Abstract—Ultrasound examination has become a vital mid-term prenatal screening method due to its numerous benefits. However, identifying fetal ultrasound sections poses significant challenges and requires the expertise of experienced doctors, especially when dealing with unfavorable fetal positions and image artifacts. The shortage of skilled doctors and the complexity of these screenings highlight the necessity of artificial intelligence to support medical professionals. Most current research targets a few specific fetal ultrasound sections for particular tasks. In reality, mid-pregnancy level II ultrasound assessments involve around ten different fetal sections. Therefore, there is a critical need for algorithms that can recognize these level II fetal sections. Furthermore, the standard ultrasound planes located in the same region exhibit a high degree of similarity. The traditional fetal slice algorithm such as ResNet generates feature maps that are sparse matrices. As a result, employing global average pooling (GAP) layers can lead to significant information loss, making it challenging to effectively differentiate between similar views. In this study, we introduce a deep-learning network designed for second-trimester anatomy ultrasound standard plane recognition. Our approach incorporates a novel spectral pooling paradigm. We introduce singular pooling to enhance information extraction from feature maps and integrate this singular pooling layer into ResNet. This methodology is tested on our dataset, which includes ten distinct standard fetal ultrasound sections from second-trimester level II examinations. Our model achieves exceptional performance, with an accuracy of 92.07% and an F1-Score of 0.919. The implementation of singular pooling significantly improves the original ResNet model. This method can effectively aid in the identification of standard plane recognition in second-trimester anatomy ultrasounds. Furthermore, we validated the efficacy of our model by applying it to two additional datasets, thereby further demonstrating its efficiency and applicability. Our code is available at <https://github.com/sysll/vectore/tree/master>

Index Terms—Level II ultrasound, image recognition, residual neural network, pooling.

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Shengjun Zhu and Jiaxin Cai are with the School of Mathematics and Statistics, Xiamen University of Technology, Xiamen 361024, China (e-mail: 19359446568@163.com; caijiaxin@xmut.edu.cn).

Runqing Xiong, Liping Zheng, and Duo Ma are with the Department of Ultrasound Imaging, Xiamen Medical College Affiliated Second Hospital, Xiamen 361021, China (e-mail: 2571186304@qq.com; 787886024@qq.com; ligangduoduo@163.com).

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I. INTRODUCTION

ULTRASOUND technology is a well-established diagnostic technique widely utilized for prenatal observation, measurement, and the diagnosis of fetal diseases [1], [2], [3], [4], [5], [6]. The acquisition of standardized ultrasound views of the fetus is a fundamental prerequisite for conducting various examinations in the domain of fetal ultrasound diagnosis. The acquisition of standardized ultrasound views of the fetus typically necessitates the expertise of skilled medical professionals and is influenced by factors such as equipment quality. Given the challenges associated with fetal plane screening and the scarcity of ultrasound doctors, the integration of standard plane recognition technology becomes imperative to mitigate the risk of misdiagnosis and enhance efficiency.

Efforts have been devoted to the classification of fetal standard ultrasound views [7], [8], [9], [10], [11], [12]. However, the current research still has limitations. Current research on standard section recognition primarily focuses on standard ultrasound (level I) examinations, which are typically conducted in low-risk pregnancies. These studies generally target the identification of three or fewer standard sections within a single body area, such as the head, face, or abdomen [13], [14], [15], [16]. There is currently no established approach for identifying multiple standard sections across different body parts. The mid-pregnancy period is crucial for anomaly screening, where accurately identifying standard fetal views is essential for detecting fetal deformities and congenital diseases. In clinical practice, second-trimester prenatal examinations require level II ultrasound assessments, which involve evaluating approximately ten standard fetal ultrasound sections, rather than just a few. Consequently, there is an urgent need to explore the classification task for ten or more views. Furthermore, views taken from the same region exhibit high similarity. The study [10] indicates that the performance of the traditional convolutional neural network (CNN) based models has significant limitations in identifying similar brain views. However, it is these highly similar views that doctors are most likely to misclassify. Therefore, developing an algorithm capable of accurately distinguishing between these views is essential. To address these challenges, we have collected a dataset of second-trimester prenatal ultrasound examinations that includes ten types of standard fetal views and propose a deep learning network called level II ultrasound

ResNet (LIUNet) to identify ten standard sections for level II ultrasound screening during mid-pregnancy.

Deep neural networks based on CNNs [17], [18], such as the ResNet model [19], have been widely applied in medical image tasks [20], [21] and have numerous excellent variants [22], [23], [24], [25] tailored for different tasks. Broadly, the ResNet model comprises two components, the feature extraction component consisting of convolutional layers that transform image information into feature maps, and the classifier components consisting of a pooling layer and a Fully Connected (FC) layer. The classifier component utilizes the information extracted from the feature maps for classification. The role of the pooling layer is to reduce the size of input data, reduce the number of features, and improve the robustness and computational efficiency. The pooling layer includes max pooling (MP), and average pooling (AP) layer and the pooling used in the classifier component of ResNet is global average pooling (GAP). However, these pooling layers usually result in an excessive loss of information [26]. ResNet integrates batch normalization (BN) [27] and Rectified Linear Unit (ReLU) [28] to efficiently extract the most significant features from the preceding feature maps. While this integration can indeed enhance the efficiency of CNN layers, it also leads to the generation of sparse feature maps. In the study [29], it was observed that as the depth of ResNet increased, the feature maps extracted by convolution became sparse, and then traditional pooling methods like GAP exhibited a decline in performance. This phenomenon occurs because there is a significant difference between the larger values and minimum value (0) within sparse feature maps and those larger values are widely scattered in their distribution. Due to its nature of calculating the average value of all elements in sparse feature maps, GAP overlooks the distribution information (the position of those larger values) and significant information (those larger values) contained in these maps. Employing GAP on these sparse feature maps can lead to the loss of crucial information. To mitigate the issue of information loss caused by the pooling layer in the classifier component, we introduce a novel approach termed spectral pooling paradigm.

Singular value decomposition (SVD) is a highly effective technique for feature extraction, commonly applied in image processing tasks [30], [31]. Prior research has indicated that spectral decomposition techniques such as truncated SVD, can be highly effective in decomposing sparse matrices, demonstrating their performance in reconstructing such matrices. Fig. 1 presents a visualization of a sparse matrix. From Fig. 1(2), it is evident that the reconstructed sparse matrix, derived from three sets of singular values and singular vectors, demonstrates the capability to capture distribution information, i.e., it can capture the positional information of non-zero values. Nevertheless, its capacity to retain significant information is limited, as certain reconstructed values are notably smaller when compared to the original values depicted in Fig. 1(1). Conversely, the matrix reconstructed utilizing five sets of singular vectors and singular values demonstrates exceptional performance in capturing distribution information and significant information, closely resembling the original sparse matrix graph. It is worth noting that the graph reconstructed

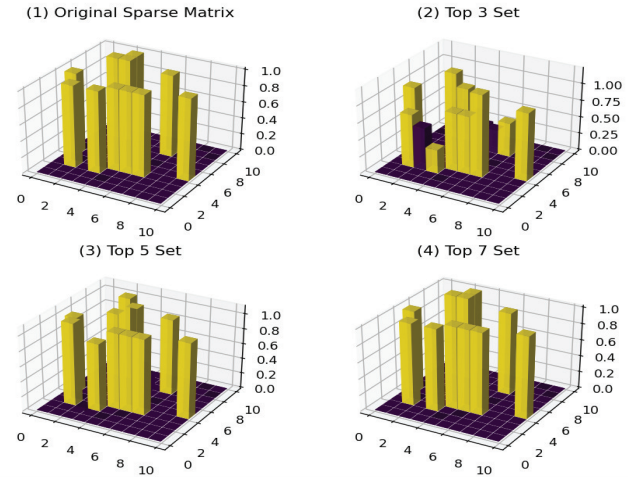


Fig. 1. A 10×10 sparse matrix and its reconstructed matrices. (1) is the original sparse matrix. (2), (3), and (4) are the reconstructed matrices using the top three, five, and seven singular values and corresponding singular vectors, respectively. The axes represent the matrix dimensions, and the bar heights indicate the matrix values.

using seven sets is not significantly different from the one reconstructed using five sets. This indicates that for each sparse feature map extracted by ResNet, only a small number of singular values and singular vectors are sufficient to reflect most of the distribution information and significant information of this sparse matrix. Therefore, we propose the matrix gene vector (MGV), a novel data structure that combines singular values and singular vectors.

The proposed matrix gene vector is formed by multiplying singular values with corresponding singular vectors and then concatenating them into a large vector. However, this data structure is distinct from typical pixel or word vector structures. Therefore, the challenge lies in effectively extracting information from the matrix gene vector. Without an efficient information extraction method, it can lead to poor performance and waste of computational resources. To tackle this challenge, we introduce the use of a one-dimensional convolutional neural network (1D-CNN), configured with a specialized hyperparameter setting method, to efficiently extract valuable information from the proposed MGV.

In this study, we focus primarily on the left singular vectors, although the same methodology can be extended to include the right singular vectors. For a feature map A , the left singular vectors can be interpreted as the directions along which the matrix AA^T undergoes stretching and scaling transformations. These vectors also encapsulate the distribution and significant information within the matrix. We propose that for similar feature map matrices A and B , after decomposition, some left singular vectors of A will closely align with certain left singular vectors of B . This similarity level is a crucial measure of image content similarity, necessitating an algorithm that can efficiently extract this information. The core function of a 1D-CNN is to assess the similarity between the convolutional kernel and the data slices. Thus, we believe that a 1D-CNN is particularly well-suited to address this challenge. Moreover, the left singular vectors of a matrix are linearly

independent, meaning that the directions they represent do not overlap, ensuring that the information contained in each vector is unique. Building on these insights, we propose two hyperparameter setting methods for using 1D-CNN to extract information from the MGv. The first method extracts information from the angles of eigenvectors, while the second focuses on the angles of feature maps.

In summary, we introduced a spectral pooling paradigm that can be integrated into deep neural networks, such as ResNet, by incorporating our proposed singular pooling layer. This approach is designed to extract more detailed image information, making it more effective for identifying fetal planes and enhancing classification performance. The spectral pooling paradigm leverages spectral components of feature maps to achieve both compression and full connectivity, enabling the reconstruction of feature maps into a linear, low-complexity structure without loss of information. In addition to our own dataset, we collected two open-source datasets for the classification task, both of which require strong image detail extraction for effective classification. Our model was comprehensively validated on all three datasets. Our contributions can be summarized as follows.

- We propose a level II ultrasound ResNet (LIUNet) for computer-aided level II ultrasound standard plane identification of second-trimester prenatal examination. We have collected a level II ultrasound standard plane dataset and demonstrated the effectiveness of LIUNet through extensive experiments. The primary objective of designing this model is to enhance its ability to extract detailed information from images and efficiently distinguish between similar views.
- We propose a spectral pooling paradigm aiming to solve the information loss problem of the pooling layer of the classifier component of deep learning neural networks.
- We propose a singular pooling layer integrating both eigenvectors and singular values in a proposed novel data structure called matrix gene vector. Two special hyperparameter setting methods are proposed for the 1D-CNN architecture of the singular pooling layer. By employing the proposed singular pooling layer in LIUNet by substituting the GAP and FC layers, we have successfully achieved computer-aided identification of standard sections for Level II ultrasound examination.

The remaining structure of this paper is as follows. Section II is the Materials. Section III is the Methodology. Section IV is the Experimental Result. Section V is the Discussion and Future Work. Section VI is the Conclusion.

II. MATERIALS

A. Our Collected Level II Ultrasound Dataset

We constructed a dataset for mid-pregnancy Level II ultrasound examinations (LIU) on fetuses. Our study received approval from the Department of Ultrasound at Xiamen Medical College Affiliated Second Hospital, China, where the data collection also took place. The Level II ultrasound examinations were performed on pregnant women aged 18 to 50, with gestational ages ranging from 22 to 26 weeks. All participants

TABLE I

THE NUMBER OF TRAINING SAMPLES AND THE NUMBER OF TEST SAMPLES FOR EACH CLASS WITHIN OUR LIU DATASET

Class	Train NO.	Test NO.	Class	TrainNO.	Test NO.
TVLV (0)	100	51	TVC (5)	133	68
TVE (1)	127	72	CVNL (6)	154	54
TVK (2)	111	37	SVP (7)	125	36
SVK (3)	147	76	TVA (8)	121	66
TVT (4)	101	43	MSVF (9)	170	77

Note: The NO. denotes the number

were single pregnancies and possessed good comprehension and expression abilities. Informed consent forms were signed by all subjects. We excluded experimental subjects with a Body Mass Index (BMI) larger than 25 kg/m^2 , an amniotic fluid depth less than 2 cm , or an amniotic fluid index less than 5 cm . The ultrasound examinations were performed using a GE E-8 advanced Color Doppler Ultrasound Machine with a convex volume probe frequency of $2\text{-}7 \text{ MHz}$ and convex probe frequency of $2\text{-}5 \text{ MHz}$, and a Samsung WS80A advanced Color Doppler Ultrasound Machine with a convex volume probe frequency of $1\text{-}8 \text{ MHz}$ and convex probe frequency of $1\text{-}7 \text{ MHz}$. The instruments were set to conditions suitable for mid-pregnancy fetal examinations. The ultrasound procedures were carried out by experienced ultrasound doctors, each with over three years of experience and prenatal training. The LIU dataset includes ten standard sections for Level II ultrasound examinations. They are Transverse view of lateral ventricle (TVLV), Transverse view of eyes (TVE), Transverse view of kidney (TVK), Sagittal view of kidney (SVK), Transverse view of thalamus (TVT), Transverse view of cerebellum (TVC), Coronal view of Nose and lips (CVNL), Sagittal view of spine (SVP), Transverse view of abdomen (TVA), Mid sagittal view of the face (MSVF). These classes are denoted as Class 0 to 9 for a 10-class classification task. The LIU dataset comprises a total of 1869 images, each with a shape of $[1, 768, 576]$. Example images of these ten categories are shown in Fig. 2(0)-(9). Data collected around January 2019 constituted the training set, while data collected around June 2019 served as the test set. Then, we adjusted all images to a size of $[1, 200, 200]$ using bi-linear interpolation before feeding them into test models. There is no overlap between the experimental subjects in the training set and those in the test set. The training set comprises 1289 images, and the test set comprises 580 images. The detailed distribution of the dataset is presented in Table I. For the training set and test set, the number of images for each category is relatively balanced.

B. Two Additional Public Datasets About Eye Disease and Skin Cancer

To evaluate our model's ability to extract detailed image information, we selected two datasets that demand high-level detail for classification.

The first public dataset is the Eye Disease Deep Learning dataset [32] which comprises 712 stained images of corneas affected by stromal ulcers. The dimensions of each image are $[3, 1592, 1728]$ pixels. The dataset includes three categories of eye disease: punctate corneal ulcer, mixed punctate stromal

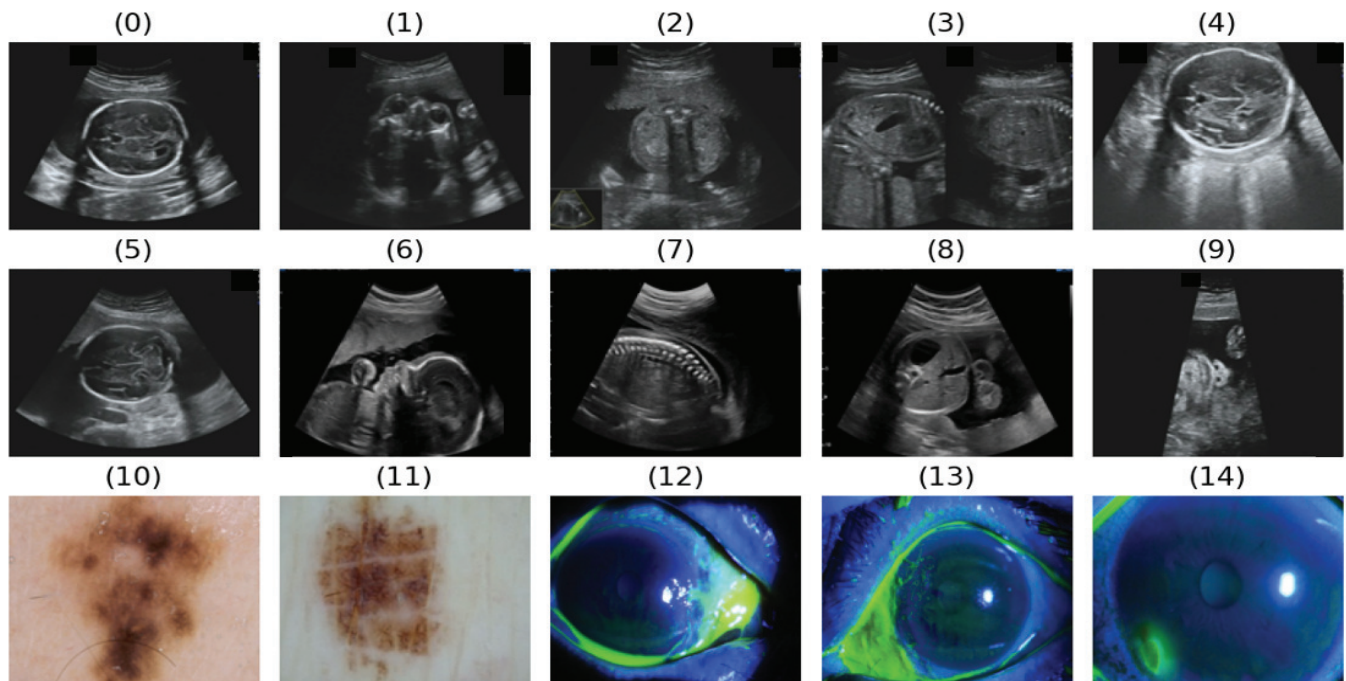


Fig. 2. The (0)-(9) show sample images from the LIU dataset. The (10) and (11) display sample images from the Skin Cancer: Malignant vs. Benign dataset. The (12), (13), and (14) present sample images from the Eye Disease Deep Learning Dataset. (0) is the transverse view of the lateral ventricle. (1) is the transverse view of the eyes. (2) is the transverse view of the kidney. (3) is the sagittal view of the kidney. (4) is the transverse view of the thalamus. (5) is the transverse view of the cerebellum. (6) is the coronal view of the nose and lips. (7) is the sagittal view of the spine. (8) is the transverse view of the abdomen. (9) is the mid-sagittal view of the face. (10) is Malignant Skin Cancer. (11) is Benign Skin Cancer. (12) is punctate corneal ulcer. (13) is mixed punctate stromal corneal ulcer. (14) is stromal corneal ulcer.

corneal ulcer, and stromal corneal ulcer, labeled as 0, 1, and 2, respectively. These categories contain 358, 263, and 91 images, respectively. Example images of these three categories are shown in Fig. 2(12)(13)(14). This dataset requires the model to effectively capture intricate texture features of the cornea, which can verify the model's ability to extract detailed information. Additionally, the imbalanced number of data categories and insufficient amount of data in this dataset can test the model's performance under conditions of imbalanced data. The second public dataset is the Skin Cancer: Malignant vs Benign dataset [33] which contains 3,297 images, each with dimensions of [3, 224, 224] pixels. The dataset is divided into two categories: Malignant Skin Cancer and Benign Skin Cancer. Images labeled as 0 represent malignant cases, with a total of 1,800 images, while those labeled as 1 indicate benign cases, with 1,497 images. Example images of these two categories are shown in Fig. 2(10)(11). This dataset necessitates the model's capacity to effectively extract detailed features from images of different skin cancers, thereby validating its aptitude for discerning meticulous information. Moreover, the dataset's balanced distribution of data categories can assess the model's performance in the context of balanced data.

For the Eye Disease dataset, we employed a 7:3 random split for training and testing. Given the specific focus on corneal injuries, which are limited to the cornea and do not involve other parts of the eye, we adopted tailored data processing techniques. The images were initially resized to [3, 400, 400] pixels using interpolation. Subsequently, we cropped the central region of each image, producing [3, 300, 300] pixel images, which were subsequently resized to [3, 200, 200]

pixels. This pre-processing step was crucial for eliminating extraneous elements like eyelashes and eyelids, thereby reducing interference and enhancing data quality. Regarding the Skin Cancer dataset, we used the partitioning provided by the official Kaggle source, where the dataset was pre-divided into training and testing sets. Subsequently, we resized all images to [3, 200, 200] using bi-linear interpolation before feeding them into the test models.

III. METHODOLOGY

A. Proposed Spectral Pooling Paradigm

Deep neural networks such as ResNet typically consist of numerous blocks, each containing several pooling layers. Therefore, the information within the extracted feature maps is generally considered to be adequately regularized. Traditionally, these models utilize a global average pooling (GAP) layer to aggregate each extracted feature map into a single average value. However, this approach leads to a significant loss of information and neglects positional details on the feature maps. As a consequence, these models such as ResNet encounter challenges in capturing intricate details of small objects in images. To address this problem, we propose spectral pooling as an alternative to the GAP layer and fully connected (FC) layer. The spectral pooling layer can be considered as an abstract pooling operation. The proposed spectral pooling paradigm employs spectral components of feature maps to achieve simultaneous feature map compression and full connection, which can reconstruct feature maps to a linear and low-computational-complexity structure without

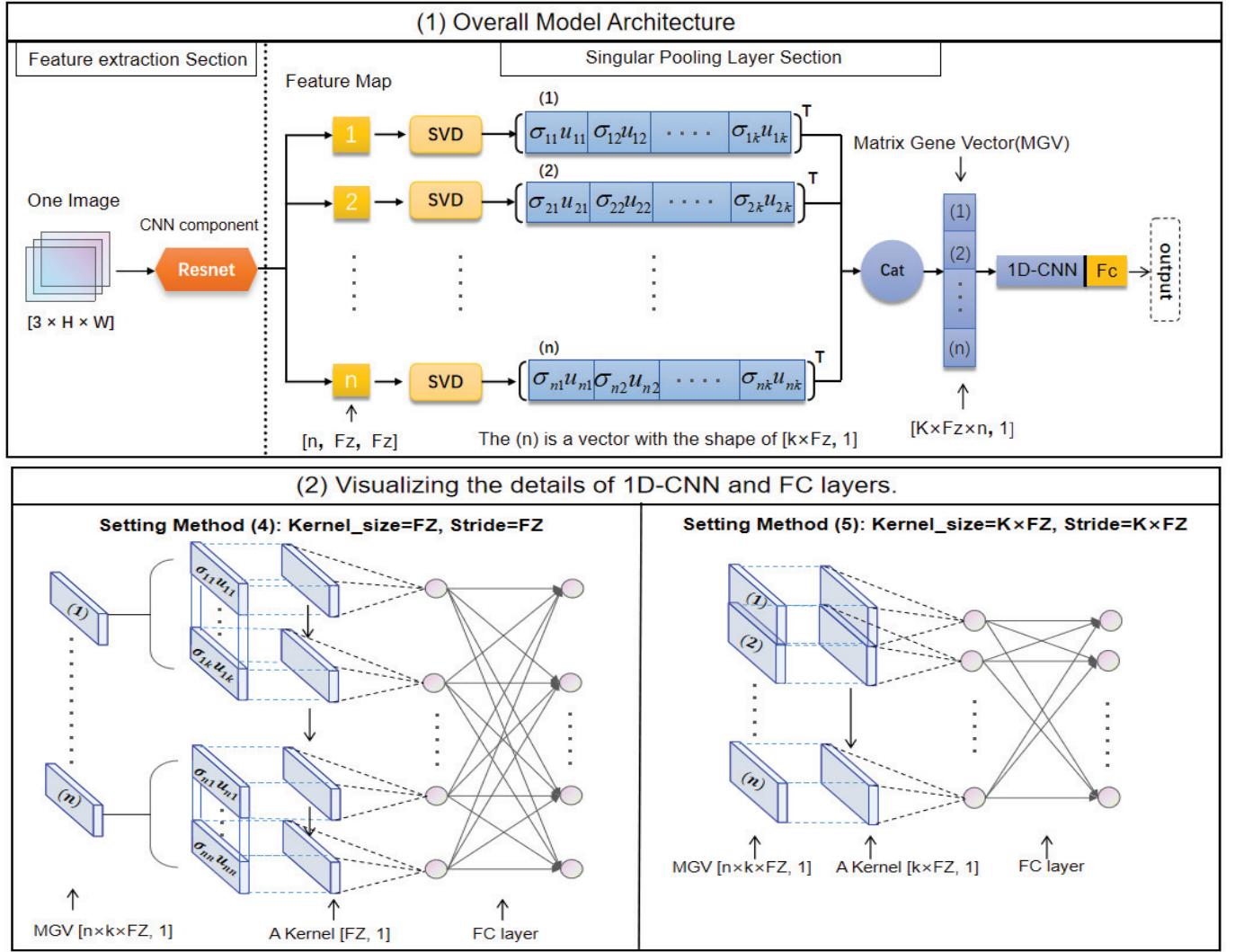


Fig. 3. (1) is the structure of our model. $[\cdot]^T$ denotes transpose. u_{ij} denotes the j -th left singular vector of the i -th feature map, and σ_{ij} denotes the j -th singular value of the i -th feature map. (2) is the visualization of 1D-CNN under two proposed CNN hyperparameter setting methods. The data shapes should be referred to as the shapes we provided in the brackets $[\cdot]$.

losing information. Input image data into a pre-trained ResNet model for feature extraction, resulting in the extraction of feature maps. We employ spectral decomposition technology, such as singular value decomposition (SVD), on the feature maps which are extracted from the deep convolutional layers of deep neural networks. Then we extract the main part of spectral components and their corresponding spectral values from each feature map, and then multiply them to concatenate a proposed matrix gene vector, which is subsequently fed into a 1D-CNN for classification.

B. Overall Framework of Proposed LIUNet With Singular Pooling

The architecture of our model is depicted in Fig. 3. Initially, we employed ResNet for pre-training on the training set. Subsequently, we freeze the CNN component of ResNet and remove the GAP and FC layers. Consequently, the ResNet model now is a feature extractor used for extracting feature maps. It will generate feature maps with dimensions of $[n, \text{feature_size}, \text{feature_size}]$, where feature_size (FZ)

denotes both the width and height of the maps and n denotes the total number of feature maps. Subsequently, we employ SVD to decompose each of these feature maps into its constituent components, namely the left singular vector matrix, the right singular vector matrix, and the singular value matrix. For each feature map, we select the top k singular values and their corresponding left singular vectors and then multiply these left singular vectors by their corresponding singular values. Finally, all the new vectors, obtained by multiplying the left singular vectors by their corresponding singular values from all feature maps, are concatenated into a large vector called the matrix gene vector (MGV). This big vector is then fed into a 1D-CNN with a special hyperparameter setting method for classification.

C. Proposed Singular Pooling

To aid understanding, we include Table II, which is a nomenclature table, to describe the important notations used in this paper.

TABLE II
THE MAIN NOTATIONS INVOLVED IN THE PAPER AND THEIR
CORRESPONDING EXPLANATIONS

Notation	Description
A	A feature map
n	The number of feature maps generated by the ResNet model
x_i	The i -th pixel row of A
v_i	The i -th right singular vector of a feature map
u_i	The i -th left singular vector of a feature map
σ_i	The i -th singular value of a feature map
λ_i	The i -th eigenvalue of a feature map
U	The left singular vector matrix
V	The right singular vector matrix
S	The singular values matrix
k	The number of singular vectors we choose for each feature map
FZ	The length of a singular vector
$\text{diag}\{\cdot\}$	A diagonal matrix with the given elements
MGV	Matrix Gene Vector
SVD	Singular Value Decomposition
MFLOPs	Million Floating Point Operations

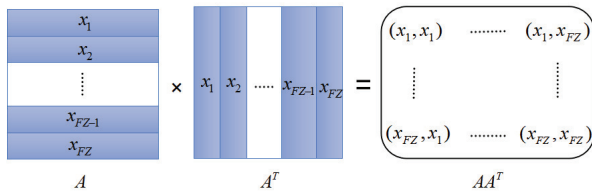


Fig. 4. Structure of AA^T . x_i signifies the i -th pixel row of A , and (x_i, x_j) denotes the dot product between x_i and x_j .

1) *SVD*: We shall commence by presenting the SVD formula which has the following general formula, $A = USV^T$. Here, A is a feature map produced by the feature extraction component of the ResNet model, U is a Left Singular Vector Matrix, V is a Right Singular Vector Matrix, and S is a Singular Values Matrix. $S = \text{diag}\{\sigma_1, \sigma_2, \dots, \sigma_{FZ}\}$, $U = [u_1, u_2, \dots, u_{FZ}]$, and $V = [v_1, v_2, \dots, v_{FZ}]$. Here, σ_i ($i = 1, \dots, FZ$) is singular value. $u_i = (u_{i1}, \dots, u_{iFZ})^T$, ($i = 1 \dots FZ$). u_i is the i -th left singular vector. $v_i = (v_{i1}, \dots, v_{iFZ})^T$, ($i = 1, \dots, FZ$). v_i is the i -th right singular vector. The FZ is the feature_size (height and width) of feature maps. On the other hand, the u_i is also the i -th eigenvector of AA^T . v_i is also the i -th eigenvector of $A^T A$. In our model, we just take U (left singular vectors) and S (singular values) into consideration. Similarly, we can also adopt right singular vectors and singular values, or consider both left singular vectors, right singular vectors, and singular values. All of the methods are interconnected. For u_i in U and σ_i in S , there exists a mathematical relationship $AA^T u_i = \lambda_i u_i$, ($i = 1 \dots FZ$, $\lambda_i = \sigma_i^2$). Here, u_i denotes the i -th eigenvector of AA^T , λ_i is i -th eigenvalue of AA^T , AA^T is the covariance matrix between the pixel rows of the feature map. It is easy to observe from Fig. 4 that it reflects similarities between pixel rows, indicating profound relationships. Simultaneously, it also preserves certain positional information from the original image matrix A .

2) *Proposed Matrix Gene Vector*: The image is fed into a pre-trained ResNet model for feature extraction, and then obtaining the feature maps. In particular, we employ SVD on

each feature map. Specifically, we extract the top k left singular vectors and their corresponding singular values from each feature map. These left singular vectors are then multiplied by their respective singular values. Furthermore, all the new vectors, obtained by multiplying the left singular vectors by their corresponding singular values from all feature maps, are concatenated into a large vector called MGV.

An in-depth explanation of MGV formation is given below. A small subset of eigenvectors and eigenvalues is often adequate to approximately reconstruct the primary information of the original image. This indicates that a small number of eigenvectors u_i and eigenvalues λ_i contain the information of the AA^T matrix. Furthermore, the eigenvector with a smaller eigenvalue typically contains less information, posing challenges for the model to effectively learn and potentially influencing the model's performance. Consequently, we opt to retain only the top k eigenvectors along with their corresponding eigenvalues.

Within the framework of the provided formula $AA^T u_i = \lambda_i u_i$, u_i is an eigenvector, signifying that the AA^T matrix undergoes stretching and scaling transformation along the direction of u_i . Therefore, eigenvectors reflect the inherent information of a matrix, and matrices that are more similar will have more similar groups of eigenvectors. Generally, the significance of an eigenvector u_i is indicated by its corresponding eigenvalue λ_i , where larger eigenvalues imply richer information content within the associated eigenvector. Therefore, the process of multiplying each eigenvector by its corresponding eigenvalue can be considered as assigning importance weights to each eigenvector. The eigenvalue λ_i is equal to the square of singular value σ_i^2 . However, given that there are significant differences in magnitudes among singular values, squaring them will further enlarge the gap, leading to excessive information reduction on some eigenvectors. While that can suppress eigenvectors with less information, our previous step of "extracting the top k singular values and corresponding singular vectors" has already ruled out those left singular vectors with less information. Therefore, our model does not need to square the singular values. Instead, we directly multiply the square root of the eigenvalues (singular value, σ_i) by corresponding eigenvectors (left singular vector).

3) *Proposed Hyperparameter Setting Methods of the Specific 1D-CNN*: The MGV is fed into a 1D-CNN model with learnable parameters for classification. The hyperparameter configuration for 1D-CNN is notably distinctive. The conventional hyperparameter setting, such as kernel_size = 3 and stride = 1, kernel_size = 5 and stride = 1 or kernel_size = 7 and stride = 1, is primarily tailored for pixel and text data. However, the input data in our study is an MGV. It is acknowledged that the traditional hyperparameter setting, with kernel size = 3 and stride = 1, is effective in extracting local information. It is, however, important to highlight the shortcomings associated with this hyperparameter setting. Firstly, it leads to overly complex computations and an excessively large CNN layer output data dimension. Secondly, in scenarios where the model encounters a situation depicted in Fig. 5(1), where a convolutional kernel spans across two eigenvectors, the convolution operation is rendered meaningless. Due to the parameter-

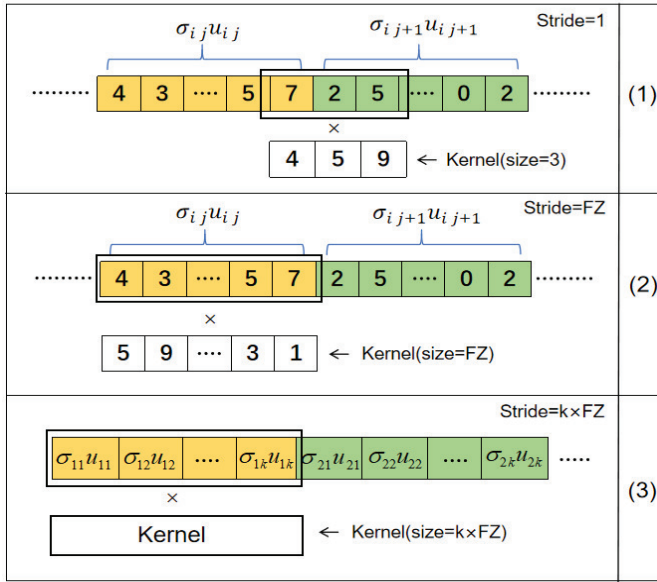


Fig. 5. CNN hyperparameter setting and analysis. We randomly generated the data inside as an example. u_{ij} denotes the j -th singular vector of the i -th feature map, and σ_{ij} represents the j -th singular value of the i -th feature map.

sharing nature of CNN kernels, this operation may affect the parameter updates of the CNN kernels, thereby necessitating an extended training period for the model to effectively filter out such information. We denote the hyperparameter setting method with kernel_size = 3 and stride = 1, kernel_size = 5 and stride = 1, and kernel_size = 7 and stride = 1 as the **Setting method (1)**, **Setting method (2)** and **Setting method (3)**, respectively. Additionally, to streamline computations and avoid scenarios like Fig. 5(1), we propose the following two alternative CNN hyperparameter setting methods.

Setting method (4). Stride = FZ, Kernel_Size = FZ. This setting method is based on convolutional operations according to the length of an eigenvector. The fundamental concept involves computing the dot product similarity between the convolutional kernel and each individual eigenvector sequentially. A higher similarity denotes a larger dot product value, signifying that the convolutional kernel exhibits a greater degree of alignment with the eigenvector. This process translates similarity information into 1D feature maps, wherein each element on the map denotes the similarity between the kernel and the corresponding position of the eigenvector. Subsequently, those feature maps are fed into the FC layer. Leveraging its robust learning capacity, the neural network can acquire intricate feature representations of the input data. This hyperparameter setting method can, to some extent, reduce 1D-CNN output data's dimension and effectively avoid situations like Fig. 5(1). The operational process is illustrated in Fig. 5(2). The visualization of the details of this hyperparameter setting method is presented in Fig. 3(2). It is undeniable that this approach also has its limitations, as it is challenging to distinguish information between feature maps from the perspective of an eigenvector.

Setting method (5). Stride = k×FZ, Kernel_Size = k×FZ. This setting method allows convolution operations to be

TABLE III
ANALYSIS OF TIME COMPLEXITY FOR 1D-CNN AND FC LAYER UNDER FIVE DIFFERENT HYPERPARAMETER SETTING METHODS

	1D-CNN	FC layer
SM (1)	$O(c_1 \times c_2 \times (L_1 - 2) \times 5)$	$O((L_1 - 2) \times c_2 \times n_{\text{class}})$
SM (2)	$O(c_1 \times c_2 \times (L_1 - 4) \times 9)$	$O((L_1 - 4) \times c_2 \times n_{\text{class}})$
SM (3)	$O(c_1 \times c_2 \times (L_1 - 6) \times 13)$	$O((L_1 - 6) \times c_2 \times n_{\text{class}})$
SM (4)	$O(c_1 \times c_2 \times L_1 \times (2 - \frac{1}{FZ}))$	$O((\frac{L_1}{FZ}) \times c_2 \times n_{\text{class}})$
SM (5)	$O(c_1 \times c_2 \times L_1 \times (2 - \frac{1}{k \times FZ}))$	$O((\frac{L_1}{k \times FZ}) \times c_2 \times n_{\text{class}})$

Note: SM denotes the hyperparameter setting method. The $O(\cdot)$ symbol denotes time complexity. FZ is the length of a singular vector. k is the number of singular vectors we choose for each feature map to construct the MGv.

performed based on the sum length of the lengths of all the top k eigenvectors in a feature map, calculating the similarity between the kernel and all the top k eigenvectors in the feature map at once. Since the top k eigenvectors of a feature map are sorted in descending order of eigenvalues, the front part of this kernel computes information corresponding to larger eigenvalues of each feature map, while the latter part does the opposite. This introduces order for each feature map's eigenvector and effectively distinguishes information between feature maps. With this configuration, CNN can operate in the form depicted in Fig. 5(3), and visualization of details of the hyperparameter configuration method is presented in Fig. 3(2). We think this hyperparameter setting method is the most recommended and rational approach.

Typically, before employing these five hyperparameter setting methods, a common practice involves using hyperparameters with stride = 1 and kernel_size = 1. This parameter configuration does not affect the matrix gene vector's structure but rather elevates its dimensionality.

D. Time Complexity Analysis

Our algorithm begins by performing SVD on the feature maps extracted from the traditional ResNet model. Then we use singular values and left singular vectors to construct the MGv, which we employ a 1D-CNN to extract information for classification. The steps involved in extracting feature maps from the ResNet model and constructing the MGv are fixed, and the complexities of the ResNet and SVD algorithms have been extensively studied. Therefore, our primary focus is on analyzing the time complexity of the 1D-CNN and the subsequent FC layer used for information extraction from the MGv under various hyperparameter settings. The 1D-CNN designed for MGv information extraction consists of two layers. The first layer features a 1D-CNN with both the kernel_size and stride set to 1. This configuration is fixed and does not require further complexity analysis. In contrast, the second layer of the 1D-CNN includes five different hyperparameter settings, which include the traditional settings methods (1), (2), and (3), as well as our proposed setting methods (4) and (5). These hyperparameter settings also influence the subsequent FC layer, making it essential to consider their effects in our analysis.

TABLE IV

THE STRUCTURE OF THE PRE-TRAINED RESNET OF OUR LIIUNET MODEL AND THE CORRESPONDING HYPERPARAMETERS FOR EACH LAYER

Layer	Architecture
Input layer	Image data (batch_size, channel, 200, 200)
Convolution	64 kernels of size 7×7, stride=2, padding=3 MaxPool2d (pool_size=3, stride=2)
1.BasicBlock	In_channels=64, out_channels=64 stride=2, block=2
2.BasicBlock	In_channels=64, out_channels=128 stride=2, block=2
3.BasicBlock	In_channels=128, out_channels=256 stride=2, block=2
4.BasicBlock	In_channels=256, out_channels=64 stride=2, block=2
GAP	
Fc	(64, n_class)

Note: n_class denotes the number of classifications. In_channels denotes the number of input channels for CNN. Out_channels denotes the number of output channels for CNN. block denotes the number of basic blocks.

TABLE V

THE STRUCTURE AND THE MAIN HYPERPARAMETERS OF 1D-CNN IN OUR LIIUNET MODEL

Layer	Architect
1.Convolution	kernel, kernel_size, stride
2.Convolution	kernel, kernel_size, stride
FC	Output=n_class

Note: The kernel denotes the number of kernels. n_class denotes the number of classes.

Since the time complexity contributed by the bias term in convolution is minimal, it can be considered negligible. For a typical 1D-CNN with input data of shape $[c_1, L_1]$, where c_1 represents the number of channels and L_1 is the length (or dimension) of the MGv, assume the kernel length is F , the stride is s , input channels is c_1 , and output channels is c_2 . The shape of its output data is $[c_2, L_2]$, where L_2 the output dimension. The time complexity for this convolution is $O(c_1 \times c_2 \times L_2 \times (2F - 1))$. The time complexity for the FC layer after 1D-CNN is $O(L_2 \times c_2 \times n_class)$, where n_class is the number of classes for classification. The detailed time complexity analysis is presented in Table III. We can observe that the time complexity of hyperparameter setting methods (5) and (4) is significantly lower than the traditional hyperparameter setting method. For the 1D-CNN module, the time complexity of hyperparameter setting methods (1), (2), and (3) follows a ratio of approximately 5:9:13, with $L_1 \gg 2, 4, 6$. The time complexity of hyperparameter setting methods (4) and (5) is close, with $FZ \gg 1$. For the FC layer, the time complexity of hyperparameter setting methods (1), (2), and (3) is close, with $L_1 \gg 2, 4, 6$. The time complexity of hyperparameter setting methods (4) and (5) follows a ratio of approximately $k:1$.

E. Implementation Details

Our code is available at <https://github.com/sysll/vectore/tree/master>. All experiments are implemented in Python 3.9 with PyTorch 2.0.1+cu117 on a computer with CPU Intel I7-12650H and GPU NVIDIA RTX2080Ti. The

architecture and hyperparameter of the pre-trained ResNet of our LIIUNET model are presented in Table IV. We utilize the traditional basic block for the fundamental structure of the ResNet convolutional layer. The number of channels increases layer by layer, starting from 64 to 128 and then to 256. This design facilitates comprehensive feature extraction during the incremental process, followed by a reduction of the extracted 256 channels to 64. This approach not only improves the model's generalization capability but also mitigates computational complexity. The structural configuration of our 1D-CNN model for the specified task is delineated in Table V.

For our LIIU Dataset, the shape of the data extracted after feature extraction by the pre-trained ResNet model is [batch_size, 64, 10, 10]. There are 64 feature maps, each with a width and height of 10. After SVD decomposition, the number of singular vectors k extracted for each feature map is 5. The length of the obtained MGv is 3200. The hyperparameters for the first layer of the 1D-CNN are stride=1, kernel_size=1, kernel=18. The hyperparameter for the second layer CNN consists of five options, (1) kernel_size=3, stride=1, and kernel=64; (2) kernel_size=5, stride=1, and kernel=64; (3) kernel_size=7, stride=1, and kernel=64; (4) kernel_size=FZ, stride=FZ, and kernel=64; (5) kernel_size= $k \times FZ$, stride= $k \times FZ$, and kernel=64. The output obtained from this layer is then fed into the FC layer to obtain the final classification results. Moreover, the learning rate for this model is set to 0.001, the optimizer chosen is Adam, the loss function employed is MSELoss, and the batch size is fixed at 100.

For the Eye Disease Deep Learning Dataset, the shape of the data extracted after feature extraction by the ResNet model is [batch_size, 64, 10, 10]. After SVD decomposition, the number of singular vectors k extracted for each feature map is 5. The length of the obtained MGv is 3200. The hyperparameters for the first layer of the 1D-CNN are stride=1, kernel_size=1, kernel=18. The hyperparameter for the second layer CNN consists of five options, setting method (1), setting method (2), setting method (3), setting method (4), and setting method (5). The output obtained from this layer is then fed into the FC layer to obtain the final classification results. The learning rate of this model is 0.001, the optimizer is Adam, the loss function is MSELoss, and the batch size is 64.

For Skin Cancer: Malignant vs Benign Dataset, the shape of the data extracted after ResNet is [batch_size, 64, 10, 10]. After SVD decomposition, the number of singular vectors k extracted for each feature map is 5. The length of the obtained MGv is 3200. The hyperparameters for the first layer of the 1D-CNN are stride=1, kernel_size=1, kernel=18. The hyperparameter for the second layer CNN consists of five options, setting method (1), setting method (2), setting method (3), setting method (4), and setting method (5). The learning rate for this model is set to 0.001, the optimizer chosen is Adam, the loss function employed is MSELoss, and the batch size is 128. The output obtained from this layer is then fed into the FC layer to obtain the final classification results.

TABLE VI

THE CLASSIFICATION PERFORMANCE COMPARISON DETAILS OF OUR MODEL AND SIX SOTA MODELS IN THREE DATASETS

LIIU				
Model	ACC	Precision	Recall	F1-Score
1. Swin-Transformer	0.1759	0.3374	0.1700	0.1297
2. ViT	0.4138	0.4254	0.4120	0.4108
3. MobileNet	0.8517	0.8665	0.8550	0.8538
4. Inception	0.8810	0.8917	0.8702	0.8793
5. VGG16	0.8190	0.8239	0.8238	0.8187
6. EfficientNet_B0	0.8569	0.8571	0.8566	0.8560
7. Ours	0.9207	0.9217	0.9150	0.9199
Eye Disease Deep Learning Dataset				
Model	ACC	Precision	Recall	F1-Score
1. Swin-Transformer	0.7238	0.7611	0.5455	0.6805
2. ViT	0.6762	0.6721	0.5085	0.6345
3. MobileNet	0.5143	0.2645	0.3333	0.3493
4. Inception	0.7714	0.7415	0.6003	0.7440
5. VGG16	0.7524	0.7305	0.6204	0.7382
6. EfficientNet_B0	0.6905	0.6099	0.5134	0.6475
7. Ours	0.7762	0.7766	0.6161	0.7485
Skin Cancer: Malignant vs Benign				
Model	ACC	Precision	Recall	F1-Score
1. Swin-Transformer	0.7773	0.8240	0.7919	0.7737
2. ViT	0.8333	0.8407	0.8378	0.8336
3. MobileNet	0.7970	0.8205	0.8067	0.7961
4. Inception	0.8591	0.8630	0.8619	0.8594
5. VGG16	0.8636	0.8640	0.8633	0.8637
6. EfficientNet_B0	0.8303	0.8344	0.8331	0.8307
7. Ours	0.8864	0.8871	0.8869	0.8865

Note: Precision and F1-Score are calculated using support-weighted mean, while Recall is calculated using micro-averaging.

IV. EXPERIMENTAL RESULTS

A. Metrics and Baselines

In order to comprehensively assess the performance of the proposed model, we employed evaluation metrics including accuracy (ACC), Precision, Recall, and F1-Score. In addition, we employed confusion matrices, Receiver Operating Characteristic (ROC) curves, and Precision-Recall (PR) plots for comparative visualization, alongside two key metrics, Area Under the Curve (AUC) and Average Precision (AP). In our comparative analysis, we have selected six state-of-the-art (SOTA) models as baselines. These encompass four well-established CNN-based models, namely VGG16 [34], EfficientNet_B0 [35], MobileNet [36], and Inception [37]. Additionally, we include two vision transformer models, specifically ViT [38] and Swin-Transformer [39]. Subsequently, the six models were applied to three datasets, and the corresponding evaluation metrics were obtained for a comprehensive comparison. It is noteworthy that, for the six SOTA models, the input image size remained consistent at [channel, 224, 224] across all three datasets.

B. Comparison With Six SOTA Models

Quantitative results of our model and six SOTA models for the three datasets are presented in Table VI. On our LIIU dataset, our model outperforms the CNN-based model by an average improvement of 6.9%. Nevertheless, we discovered that the two transformer-based models, ViT and Swin-Transformer, exhibited inferior performance. We attribute this to the limited number of images per class in

LIIU, with approximately 100 images per class. Furthermore, we observed a normal reduction in loss values and the transformer-based model exhibited approximately 10 times more parameters than the convolution-based model. Consequently, we hypothesize that the poor performance of the ViT and Swin-Transformer models is attributable to their complexity and the excessive number of parameters, resulting in overfitting when dealing with a relatively small dataset. For the Eye Disease Deep Learning Dataset, the dataset is relatively small, comprising 712 images, and exhibits a significant imbalance in the number of images among the three categories in the training set. Despite these challenges, our model also achieves high accuracy (ACC = 0.7768). Notably, our model surpasses CNN-based models on average by 3.4% in accuracy and outperforms vision transformer models on average by 7.1%, highlighting the efficacy of our proposed model. Concerning the Skin Cancer: Malignant vs Benign dataset, the training set comprises a substantial number of images depicting both benign and malignant tumors, with a well-balanced distribution between the two classes. A larger dataset, characterized by an increased number of training samples, facilitates enhanced learning of feature representations by neural network models. Consequently, all six models in our study exhibit commendable performance. On average, our models surpass two vision transformer models by 7.8% in terms of accuracy and outperform four CNN-based models by an average of 4.58%. Notably, our model achieves the highest performance in our experiments.

We believe that Transformer-based models primarily capture global information, which is not essential for our task. Instead, our classification task requires capturing local texture details. CNN-based models demonstrate superior performance at extracting local information and are more robust in capturing fine details, which explains why they outperform Transformer-based models across our three datasets. Notably, the Inception model performs relatively well by using multiple convolutional kernels to extract detailed features. However, many CNN-based models still struggle to distinguish highly similar planes. In our model, we replaced ResNet's GAP and FC layer with a singular pooling layer to enhance the extraction of sparse matrix information and preserve more detailed features, leading to the best overall performance.

C. Ablation Studies

In this section, we conducted a series of ablation studies to evaluate the effectiveness of the singular pooling module and validate the effectiveness of the proposed two hyperparameter configurations for the 1D-CNN.

1) *Effectiveness of Singular Pooling*: This experiment was conducted with the primary aim of illustrating the effectiveness of the singular pooling module. We began by training a ResNet model and recording its performance results. Following this, we froze the parameters of the CNN segment within ResNet and replaced the GAP and FC layers with the singular pooling layer. The model was then trained exclusively on the singular pooling layer, updating only its parameters. This approach provided a robust benchmark, allowing us to compare the original ResNet model's performance, which utilized the GAP and FC

TABLE VII

THE CLASSIFICATION PERFORMANCE COMPARISON DETAILS OF OUR MODELS WITH VARIED HYPERPARAMETERS SETTING METHODS IN THREE DATASETS

LIU						
Model	ACC	Precision	Recall	F1-score	SM	p
ResNet	0.8724	0.8920	0.8697	0.8749	/	/
Ours	0.9207	0.9217	0.9150	0.9199	(5)	2×10^{-16}
Ours*	0.9138	0.9144	0.9068	0.9130	(4)	9×10^{-16}
Ours*	0.5345	0.5520	0.5484	0.4672	(3)	0.99999
Ours*	0.6586	0.6218	0.6142	0.6187	(2)	0.99998
Ours*	0.9138	0.9159	0.9092	0.9130	(1)	1×10^{-10}

Eye Disease Deep Learning Dataset						
Model	ACC	Precision	Recall	F1-score	SM	p
ResNet	0.7476	0.7367	0.5954	0.7214	/	/
Ours	0.7762	0.7766	0.6161	0.7485	(5)	1×10^{-3}
Ours	0.7714	0.7933	0.6106	0.7420	(4)	5×10^{-1}
Ours*	0.3762	0.1689	0.3690	0.2222	(3)	0.99999
Ours*	0.7476	0.6700	0.5653	0.7039	(2)	9×10^{-1}
Ours*	0.7524	0.6650	0.5635	0.7058	(1)	7×10^{-1}

Skin Cancer: Malignant vs Benign						
Model	ACC	Precision	Recall	F1-score	SM	p
ResNet	0.8712	0.8738	0.8733	0.8715	/	/
Ours	0.8864	0.8871	0.8869	0.8865	(5)	1×10^{-6}
Ours	0.8818	0.8819	0.8811	0.8818	(4)	9×10^{-7}
Ours*	0.8758	0.8760	0.8753	0.8758	(3)	8×10^{-1}
Ours*	0.8803	0.8816	0.8814	0.8805	(2)	1×10^{-1}
Ours*	0.8833	0.8836	0.8833	0.8834	(1)	8×10^{-4}

Precision and F1-Score are calculated using support-weighted mean, while Recall is calculated using micro-averaging. Ours*, a set of comparative models, is designed to demonstrate the significant limitations of the traditional parameter setting method. SM is the hyperparameter setting method, and p is the p-value. We believe that the performance of the model will approximately follow a normal distribution after continuous training. Therefore, after the model's performance stabilizes, we select a continuous range of training epochs and their corresponding accuracy data. This set of data will be used to perform a One-Sample t-test against the best ACC of the traditional ResNet to calculate the p-value. We think that if $p < 0.05$, it indicates that the performance of our model is significantly higher than the traditional model, which is statistically significant.

modules, with the results obtained after integrating the singular pooling layer. Table VII offers a detailed comparison between the ResNet model and our models employing five distinct 1D-CNN hyperparameter settings. The results reveal a significant superiority of our model across all three datasets. Specifically, on the Level II ultrasound dataset, our model achieved a 4.83% improvement. On the Eye Disease Deep Learning Dataset, there was a 2.86% enhancement, and on the Skin Cancer: Malignant vs. Benign training set, our model demonstrated a 1.52% improvement. To validate the superiority of our model and the efficiency of our hyperparameter setting method, we conducted a one-tailed test. This statistical analysis confirmed that our algorithm's enhanced performance over traditional ResNet was not due to chance. Concretely, on LIU dataset, our model with our proposed setting methods gets a p-value of approximately 10^{-16} , significantly surpassing the performance of ResNet. Similar results were observed in the Skin Cancer: Malignant vs Benign dataset. While the Eye Disease dataset showed a higher p-value due to data imbalance, our model with setting method (5) still performed exceptionally well.

To visually compare the effects, we generated confusion matrices to illustrate the performance of our model in comparison to the ResNet model across various datasets, as illustrated in Fig. 6. For the LIU dataset, while the original ResNet

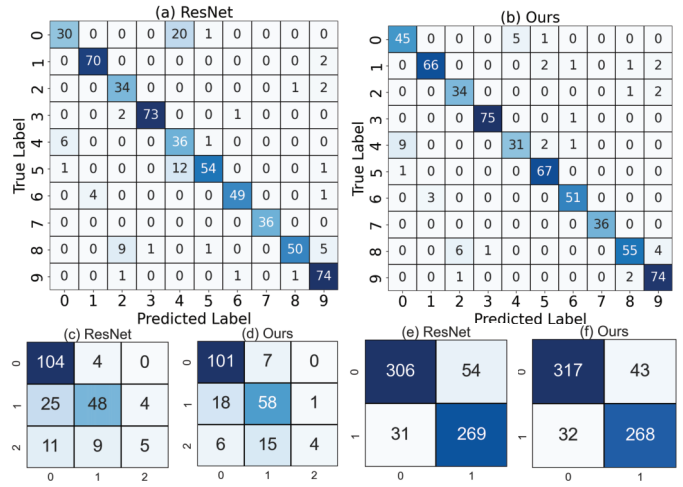


Fig. 6. Comparison of the Confusion Matrices obtained by the ResNet Model and Our Model using hyperparameter setting method (5). (a) and (b) are the confusion matrices of the ResNet and our model for the LIU dataset. (c) and (d) are the confusion matrices of the ResNet and our model for the Eye Disease Deep Learning dataset. (e) and (f) are the confusion matrices of the ResNet and our model for the Skin Cancer: Malignant vs Benign dataset. The vertical axis represents True Labels and the horizontal axis represents Predicted Labels.



Fig. 7. The three difficult-to-differentiate standard views. (1) is a transverse view of the lateral ventricle, (2) is a transverse view of the thalamus, and (3) is a transverse view of the cerebellum.

model performs well overall, it struggles with distinguishing between classes 0 (lateral ventricle cross-section), 4 (thalamus cross-section), and 5 (cerebellum cross-section). Specifically, the model frequently misclassifies images from classes 0 and 5 as class 4. This challenge in distinguishing these three planes arises primarily from their proximity within the fetal head region, as depicted in Fig. 7. Anatomically, the head region is dense with intricate details, which are further obscured by the inherent fuzziness of ultrasound images. As a result, accurately distinguishing these planes requires models capable of extracting fine details. The traditional GAP layer struggles with significant information loss during detail extraction, which hampers its performance in distinguishing these specific categories. In contrast, our singular pooling layer effectively captures complex details within the images, enhancing the model's ability to differentiate these three planes while maintaining high prediction accuracy for other planes. By replacing the GAP and FC layers with the singular pooling layer, our model not only preserved its overall classification performance but also achieved substantial improvements in distinguishing these critical categories. This comparison highlights the effectiveness of the singular pooling layer in capturing complex details within the images.

Moreover, our model demonstrates strong performance on two public datasets. Specifically, our model significantly enhances the prediction accuracy for class 1 in the Eye Disease Deep Learning Dataset. Regarding the skin cancer: malignant vs. benign dataset, compared to the original ResNet model, our model demonstrates a notable improvement in predicting class 0. Overall, our model with singular pooling far outperforms the ResNet model without singular pooling.

2) *Effectiveness of 1D-CNN Hyperparameters:* In this experiment, we conducted an ablation study to assess the efficacy of the proposed two CNN hyperparameter settings method. Our model was evaluated in five hyperparameter setting methods, (1) Stride =1, Kernel_size =3; (2) Stride =1, Kernel_size =5; (3) Stride =1, Kernel_size =7; (4) Stride = FZ, Kernel_size = FZ; (5) Stride = $k \times \text{FZ}$, Kernel_size = $k \times \text{FZ}$. The top three setting methods are conventional approaches commonly used in image and text data processing. In contrast, the fourth and fifth hyperparameter setting methods were developed through our in-depth analysis. The outcomes corresponding to these five hyperparameter settings methods are presented in Table VII. Notably, our model consistently achieved the best performance with hyperparameter setting method (5) and (4), whereas methods (3), (2) and (1) exhibited relatively inferior performance. Specifically, the performance of each setting method shows a clear trend, with performance ranked from best to worst as follows: (5), (4), (1), (2), (3).

To further validate the performance differences among these five settings, we plotted the relationship between epochs and accuracy (ACC) in Fig. 8. The dashed line in the figure indicates the performance of the traditional ResNet model on this dataset. It is evident that methods (5) and (4) require only 1-2 epochs of training to achieve competitive and superior results, whereas method (1), (2) and (3) demands more epochs to filter out irrelevant information. Furthermore, we can observe a clear layering phenomenon, with hyperparameter setting method (5) consistently at the top, followed by methods (4), (1), (2), and (3) in order. This phenomenon is most evident in our dataset and the Eye Disease Deep Learning dataset, but is less pronounced in the Skin Cancer: Malignant vs. Benign dataset. These results further underscore the effectiveness of the proposed approaches discussed in Section III-C.3. We believe that the performance differences among hyperparameter setting methods (1), (2), and (3) can be attributed to the quantity of ineffective information. As illustrated in Fig. 9, for the hyperparameter setting method (1), when the convolutional kernel crosses two singular vectors, the similarity information becomes meaningless. However, due to its smaller kernel size, it can effectively bypass this ineffective convolution operation with just two additional steps. Consequently, it generates relatively less ineffective information and exerts minimal interference on the model. In contrast, as shown in Fig. 9(2) and (3), hyperparameter settings (2) and (3) utilize larger convolutional kernels, necessitating four and six additional steps, respectively, to escape these meaningless computations. This results in a greater accumulation of ineffective information. Due to the parameter-sharing characteristic of convolutional kernels, this ineffective

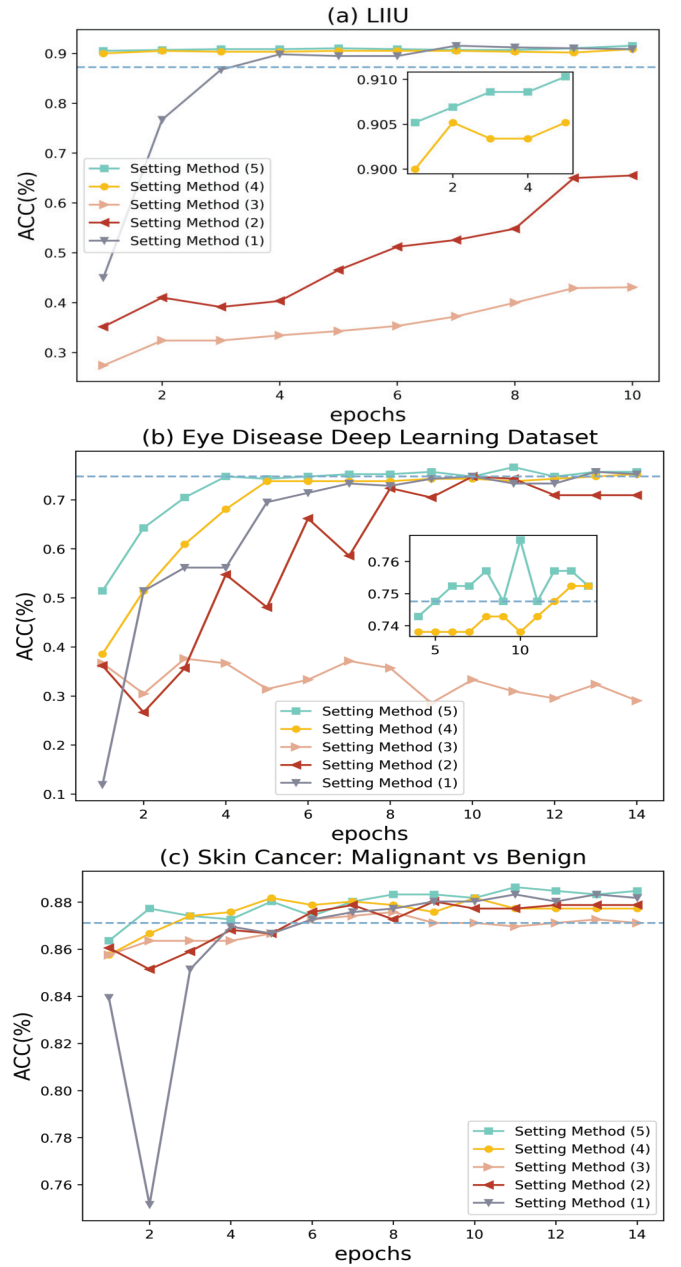


Fig. 8. The relationship between epochs and accuracy (ACC) for models with different hyperparameter settings across three datasets is illustrated, with (a) showing results for the LIU dataset, (b) for the Eye Disease Deep Learning dataset, and (c) for the Skin Cancer: Malignant vs. Benign dataset.

information will participate in parameter updates, affecting model performance. So, more training is necessary for the model to filter out ineffective information. Furthermore, this ineffective information can mislead the model and ultimately degrade its performance. The hyperparameter setting method (5) fully utilizes the information from the MGVS, yielding the best results from the model. Additionally, the 1D-CNN with setting method (5) generated the smallest feature maps with this parameter setting, which significantly reduces resource requirements.

To provide a practical and intuitive assessment of the complexity of our model under various hyperparameter settings in

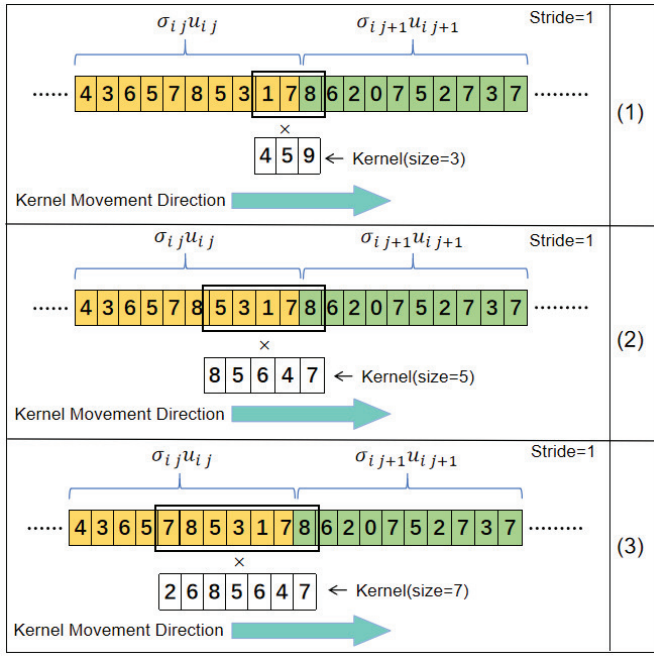


Fig. 9. Analysis diagram of 1D-CNN under traditional hyperparameter settings. u_{ij} denotes the j -th left singular vector of the i -th feature map, and σ_{ij} represents the j -th singular value of the i -th feature map.

TABLE VIII

ANALYSIS OF MFLOPS FOR 1D-CNN AND FC LAYERS UNDER FIVE HYPERPARAMETER SETTINGS

	SM (1)	SM (2)	SM (3)	SM (4)	SM (5)
1D-CNN	6.216	10.355	14.487	2.073	2.073
FC layer	1.151	1.150	1.149	0.115	0.023

Note: SM denotes the hyperparameter setting method.

our LIU dataset, we present Table VIII. This table presents the Million Floating Point Operations (MFLOPs) of the second-layer 1D-CNN and the subsequent FC layer across different hyperparameter setting methods. We can observe that the actual complexity aligns well with the theoretical derivations presented in subsection III-D. This result demonstrates the accuracy of our theoretical derivations. The intuitive numerical comparison further highlights the superior performance of the hyperparameter setting method we proposed.

D. Analysis of the Model Robustness to Ultrasound Images

Ultrasound images have inherent limitations, such as speckle noise, blurred boundaries, and image artifacts. To thoroughly analyze our model, our ultrasound specialists selected four test images that were significantly affected by these limitations, and we then conducted an in-depth analysis using these images. Our analysis results are shown in Fig. 11. Subfigures (a), (b), (c), and (d) correspond to categories 0, 4, 5, and 8, respectively, where the model's prediction performance was relatively lower, making them ideal for evaluation. Each of these images exhibits some form of artifact. (a) has blurred boundaries on both sides, (b) lacks clear internal texture, and (c) and (d) contain substantial variations in brightness due to speckle noise. Additionally, (d) has most of its boundaries

TABLE IX

A DETAILED COMPARISON OF THE CLASSIFICATION PERFORMANCE OF OUR MODEL ACROSS VARIOUS CATEGORIES IN THE LIU DATASET

class	precision	recall	f1-score
TVLV (0)	0.8182	0.8824	0.8491
TVE (1)	0.9565	0.9167	0.9362
TVK (2)	0.8293	0.9189	0.8718
SVK (3)	0.9868	0.9868	0.9868
TVT (4)	0.8611	0.7209	0.7848
TVC (5)	0.9306	0.9853	0.9571
CVNL (6)	0.9444	0.9444	0.9444
SVP (7)	1.0000	1.0000	1.0000
TVA (8)	0.9322	0.8333	0.88
MSVF (9)	0.9024	0.961	0.9308

obscured by surrounding tissue, further increasing blurriness. To visually assess the impact of these limitations, we compared the prediction scores of our model and the traditional ResNet on these four images. Robustness is typically reflected in a model's stability against input perturbations. If a model assigns a high prediction score to the correct category despite noise, it demonstrates stronger confidence in its classification and greater resilience to interference. From Fig. 11, it can be observed that our model consistently achieves higher prediction scores for the correct categories compared to the traditional ResNet. Notably, our model correctly classified image (a), while ResNet misclassified it due to the influence of blurred boundaries and artifacts. These findings indicate that our model demonstrates enhanced robustness in addressing the limitations of ultrasound images.

V. DISCUSSION AND FUTURE WORK

A. Clinical Contributions

In clinical practice, sonographers often face challenges in accurately identifying three highly similar transverse views during fetal examinations: the transverse view of the lateral ventricle (class 0), the transverse view of the thalamus (class 4), and the transverse view of the cerebellum (class 5). The morphological resemblance of these cross-sections can lead to misjudgments, especially when physicians experience visual fatigue during extensive screenings. Our model addresses this challenge by demonstrating superior performance in identifying level II ultrasound fetal views across ten categories, which are crucial for comprehensive fetal condition assessments that typically require multiple views. We introduced a singular pooling layer to replace the traditional GAP and FC layers in our experiments. This modification allowed our model to capture fine-grained image information more effectively, enhancing its ability to discriminate between the transverse view of the lateral ventricle, the transverse view of the thalamus, and the transverse view of the cerebellum. In particular, we observed a significant improvement in the identification of the transverse view of the cerebellum, bringing its accuracy closer to clinical application standards. Our model can serve as a preliminary diagnostic tool, reducing the workload for physicians and improving classification efficiency. Physicians would only need to verify the model's output against the images, conducting secondary screenings for

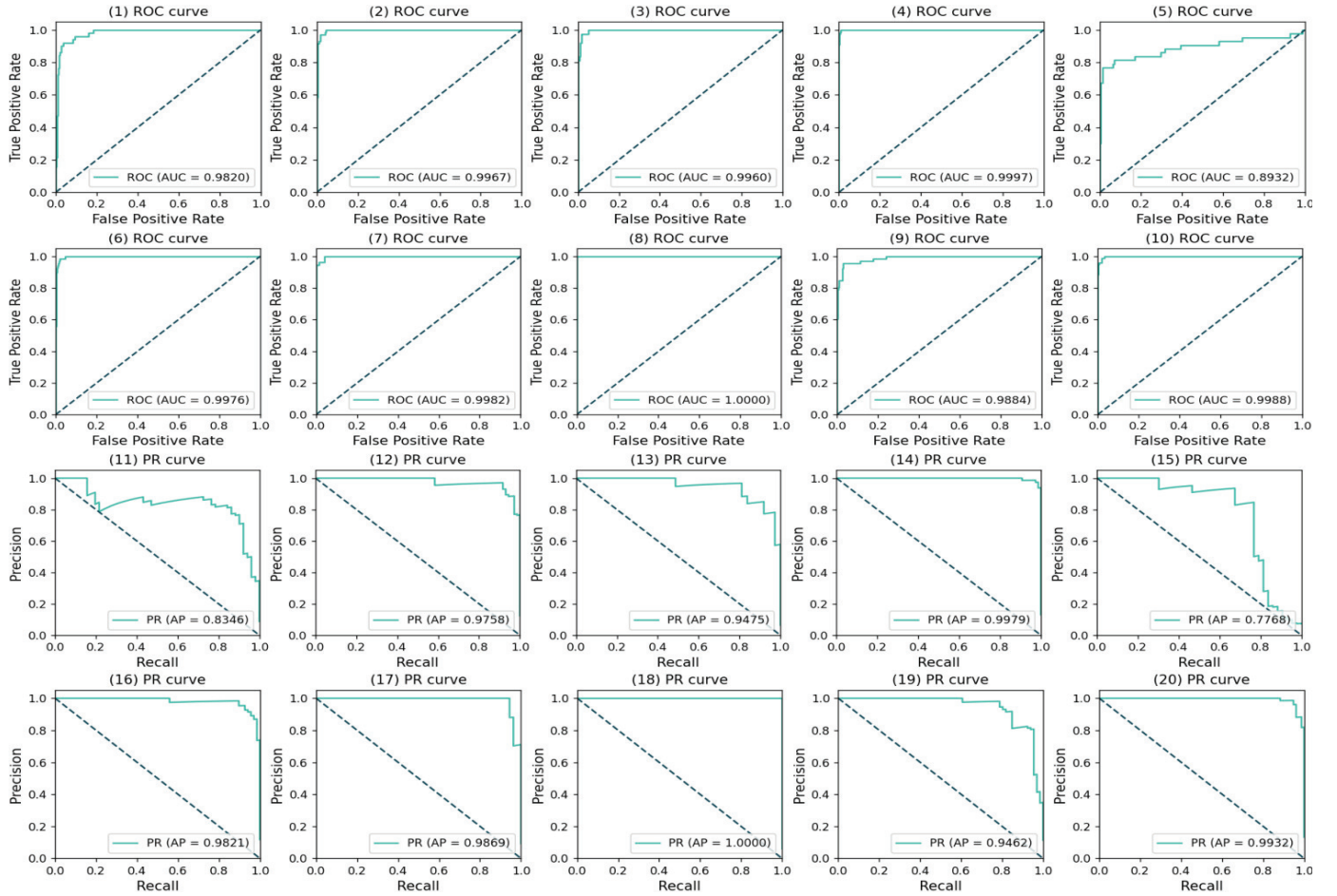


Fig. 10. The ROC and PR curves of our model in ten classes. Subfigures (1) to (10) illustrate the ROC curves for Classes 0 to 9, while Subfigures (11) to (20) depict the corresponding PR curves for these same classes.

a few inconsistent samples. This process effectively streamlines the diagnostic workflow. As shown in Table IX, our model achieved a recall and precision of over 90% in most view classification tasks. It reached 100% recall and 100% precision in class 7 (sagittal view of the spine) and nearly 100% in class 3 (sagittal view of the kidney). This high accuracy allows physicians to allocate less focus to these views during secondary screenings, thereby narrowing the range of categorization and utilizing their resources more efficiently. Additionally, the performance of our model in identifying the transverse view of the cerebellum is notably high. Fig. 10 presents the ROC (Receiver Operating Characteristic) curves and PR (Precision-Recall) curves for our model's predictions across ten categories. A higher AUC (Area Under the Curve) value indicates better performance in distinguishing between positive and negative samples. Simultaneously, a higher area under the PR curve suggests the model can recall more positive samples while maintaining a high precision rate. Our model achieves AUC and area values greater than 0.9 for most categories, strongly demonstrating its excellent classification ability and generalization performance. This provides robust support for applying our model in real-world scenarios.

Our study differs from existing research in several significant ways. The study [10] collected the largest fetal ultrasound planes dataset at the time and conducted a

comprehensive evaluation of various traditional CNN-based models. They observed that while CNN-based models demonstrate remarkable performance in differentiating between various fetal ultrasound planes, their capacity to differentiate between different brain views is comparatively limited. Different from this study, our study enhances the CNN-based ResNet model, markedly improving its ability to identify different brain views. Notably, our model's performance in recognizing the transverse view of the cerebellum nearly meets clinical standards. The study [16] employed a neural network to identify cardiac views and distinguish between normal hearts and complex congenital heart disease. In contrast to this study, our research is focused on the classification of 10 distinct ultrasound planes. Furthermore, conventional second-trimester ultrasound assessments typically necessitate the examination of a minimum of 10 distinct planes, encompassing diverse regions. The study [8] assessed the feasibility of employing machine learning algorithms to classify first-trimester fetal brain ultrasound views. In contrast, our study focuses on second-trimester fetal ultrasound views, which encompass a range of regions and categories of views.

B. Algorithm Contributions

We introduced a novel singular pooling layer to supplant conventional GAP and FC. This layer serves the purpose

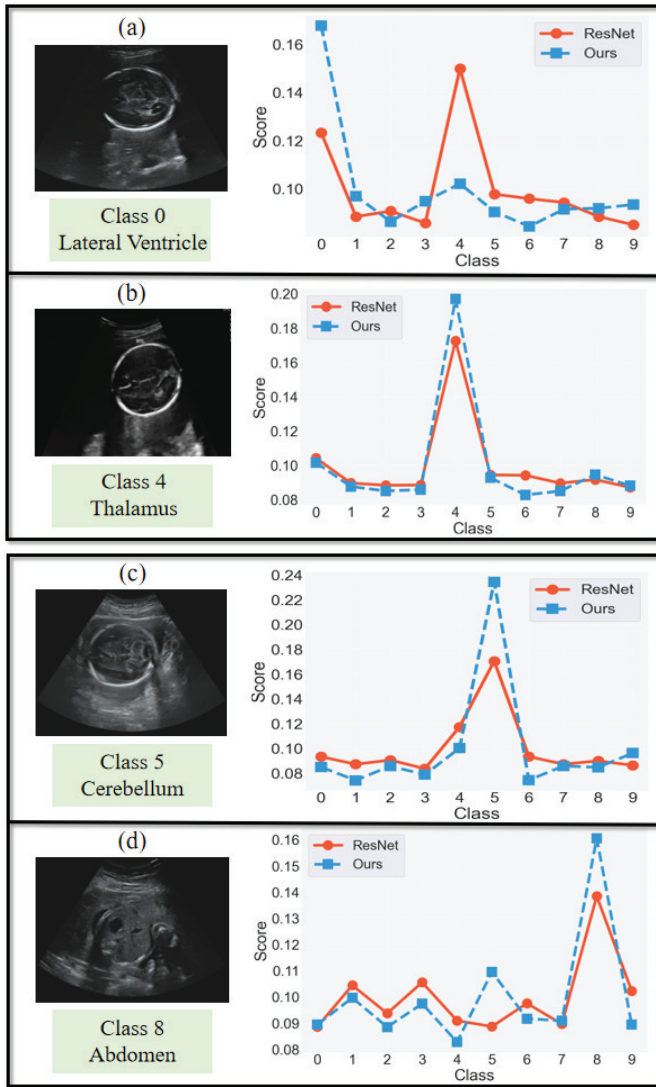


Fig. 11. Comparison of prediction scores between our model and the traditional ResNet model on ultrasound images with significant limitations. (a) is the transverse view of the lateral ventricle. (b) is the transverse view of the thalamus. (c) is the transverse view of the cerebellum. (d) is the transverse view of the abdomen.

of extracting deeper-level information within feature maps, specifically tailored for fetal ultrasound image classification. We devised a hybrid approach, termed “matrix gene vector”, incorporating both singular values and singular vectors. Additionally, we implemented several specialized CNN parameter settings for these matrix gene vectors, effectively reducing the model’s parameters while substantially enhancing its performance. Our research has opened avenues for future investigations. The matrix gene vector we proposed is a special method that can convert two-dimensional data into one-dimensional data while preserving a substantial amount of information, and this process does not rely on learnable parameters to extract useful information. Previous research [26] has proposed Singular Value Decomposition (SVD) pooling to mitigate the information loss typically associated with pooling layers. However, this approach only involves the maximum singular value. In contrast, our work considers both singular

values and singular vectors to be essential components of the reconstructed matrix. Singular values contain minimal information, while the primary information after SVD resides in the singular vectors. The singular values indicate the level of importance of these singular vectors. Unlike the study [26], our study introduces a singular pooling layer that integrates both singular values and vectors, aiming to extract information from the final feature maps comprehensively. Prior studies [23], [24] primarily focused on enhancing the feature extraction components of ResNet models, emphasizing improvements to the structure of basic blocks and residual mappings to increase the representational capacity of the extracted feature maps. In contrast, our research focuses on the classifier component, specifically targeting the GAP and FC layers, to enable more effective information extraction from feature maps. Additionally, while previous works [30], [31] utilized SVD for image processing by reconstructing images into new two-dimensional matrices, our approach employs SVD to transform feature maps into a unique one-dimensional data structure. Our research presents a paradigm for reducing two-dimensional data to a one-dimensional representation while preserving a substantial amount of information. This approach can be generalized to other spectral decomposition techniques.

Furthermore, our model demonstrates notable distinctions from previous models in the fetal planes classification task. The study [7] utilized a CNN-based stacking ensemble method for the classification of fetal ultrasound planes. The method used three pre-trained CNN-based models, AlexNet, VGG-19, and DarkNet-19, to extract features. These features were then fed into random forest classifiers to make predictions. This study focuses on the diversity of feature maps extracted by various models. Different from this study, our study focuses on improving the ability of feature extraction from a model to differentiate standard fetal ultrasound planes more effectively. The study [40] introduced a texture feature fusion method for classifying fetal facial ultrasound planes. The approach began with the extraction of texture features, including Local Binary Patterns and Histograms of Oriented Gradients, which were then fused before applying a support vector machine (SVM) for classification. This study emphasizes image processing methods to extract detailed features. In contrast, our research focuses on enhancing the classification model’s inherent ability to extract detailed information from images.

C. Drawbacks

This study has several limitations. Although our model has improved in differentiating the transverse view of the thalamus, its performance still falls short of clinical requirements. Therefore, physicians need to pay more attention to this view during secondary screenings. The dataset we used in the study is relatively small. If the dataset were larger, the model would be able to learn more abundant information. Although our model effectively extracts significant information and distribution information from sparse matrices, improving its ability to distinguish between similar images, there is a risk of overfitting when the task is simple. However, we believe that it can help reduce this risk by reducing the value of hyperparameter k . The impact of the magnitude

differences between the singular values obtained after SVD on our model was not fully considered in our research. Significant discrepancies among the singular values after SVD may lead to large differences in magnitude among different data segments on MGv, thereby adversely affecting the operations of 1D-CNN.

D. Future Work

Our research agenda will include a comprehensive investigation into 1D-CNN architectures designed to enhance the extraction of information from matrix gene vectors. Our objective is to extend the utilization of matrix gene vectors to encompass a diverse range of models in future research endeavors. Additionally, we explored several optimization algorithms [41], [42], [43] that can be utilized for hyperparameter selection. The study [41] proposed improving the computational efficiency of an incomplete data analysis model by applying an enhanced Particle Swarm Optimization (PSO) algorithm for hyperparameter optimization. The study [42] introduced a novel Hybrid-of-Evolutionary-Schemes (HOEs) algorithm for high-accuracy calibration of industrial robots. The study [43] suggested utilizing diversified regularization techniques based on the Levenberg-Marquardt algorithm to improve training for more effective manipulator calibration. We believe that, with appropriate adjustments, these methods can effectively identify better 1D-CNN hyperparameter settings of our model. Additionally, models such as VGG utilize a convolution and ReLU structure without batch normalization, which may still introduce some degree of sparsity. Meanwhile, variants like MobileNet and ShuffleNet incorporate batch normalization with ReLU, but their convolutional layers have distinct characteristics. Therefore, future research should further investigate their sparsity and evaluate the potential of singular pooling layers in these architectures.

VI. CONCLUSION

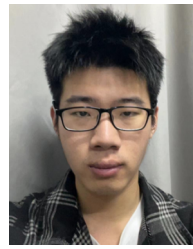
In this study, we introduced a singular pooling layer to replace the GAP and FC layers in the ResNet model, significantly enhancing the model's ability to extract detailed image features. Initially, we applied Singular Value Decomposition (SVD) to each feature map derived from ResNet's CNN component. Then, we reconstructed the feature maps into a linear structure with low computational complexity, referred to as MGv. Subsequently, we utilized a 1D-CNN with two specialized hyperparameter setting methods to decode MGv's data structure for classification. By integrating both singular vectors and singular values, our proposed model significantly improves feature extraction capabilities. We evaluated our model's ability to extract detailed features using a dataset we collected, which included ten distinct standard fetal ultrasound sections from second-trimester level II examinations, as well as two public datasets. These datasets demand models with exceptional feature extraction capabilities. The experimental results indicate that our model surpasses state-of-the-art methods. Furthermore, in our dataset, our model achieved an improvement of 4.8%, with a high accuracy of 92.07%, notably enhancing the performance of distinguishing the transverse views of the lateral ventricle, thalamus, and cerebellum.

This improvement is crucial for assisting sonographers in identifying level II fetal sections. Additionally, in two publication datasets, our model also achieved an improvement of about 1.52% and 2.86% compared to the original ResNet model, which suggests the effectiveness of our model.

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Shengjun Zhu (Student Member, IEEE) is currently pursuing the bachelor's degree in information and computational science with Xiamen University of Technology. His research interests include computer vision, deep learning, and medical image processing.



Jiixin Cai received the B.Sc. and M.S. degrees in bio-medical engineering from Southern Medical University, in 2008 and 2011, respectively, and the Ph.D. degree in mathematics (information and computational science) from Sun Yat-sen University in 2014. He worked as a Lecturer with Xiamen University of Technology from 2014 to 2023. He has been an Associate Professor with Xiamen University of Technology since 2023. He has published over 38 peer-reviewed articles. His current research interests include medical informatics and machine learning.



Runqing Xiong is currently the Director, the Chief Physician, and a Professor with the Department of Ultrasound Imaging, Xiamen Medical College Affiliated Second Hospital. She has five years of experience in surgery and has been working in ultrasound imaging for over 20 years, specializing in ultrasound diagnosis of abdomen, heart, superficial organs, large blood vessels, fetal heart, and other areas. She is a member of the Ultrasound Branch of Fujian Medical Association and the Professional Committee of the Cross Strait Medical and Health Exchange Association. She is the Director of Fujian Ultrasound Medical Engineering Society. She is the Vice Chairperson of the Ultrasound Branch of Xiamen Medical Association. She is a Key Talent introduced by Xiamen and the "March 8th Red Flag Bearer" of Xiamen.



Liping Zheng received the master's degree in medicine from Fujian Medical University. She is currently a Physician with the Department of Ultrasound Imaging, Xiamen Medical College Affiliated Second Hospital.



Duo Ma is currently the Deputy Director and the Deputy Chief Physician with the Department of Ultrasound Imaging, Xiamen Medical College Affiliated Second Hospital. She has been working in ultrasound imaging for over 18 years. She is a member of the Ultrasound Branch of Fujian Strait Medical Exchange Association, the Ultrasound Society of Xiamen Medical Association, and the Ultrasound Branch of Xiamen Physician Association. She is the Director of Xiamen Ultrasound Medical Engineering Society.